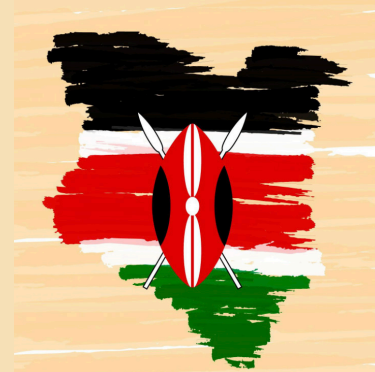


Summary of Key Research Findings for Dissemination

Overview

A series of recent studies undertaken through collaborative efforts involving the National Tuberculosis, Leprosy and Lung Disease Program, research partners, and implementing agencies has generated important evidence to strengthen tuberculosis prevention, diagnosis, treatment, and care in Kenya. Together, these studies provide complementary insights on service delivery gaps, stigma, rights and gender barriers, and public understanding of TB.



The Studies

01

The **first study** examined barriers and facilitators across the TB care cascade among adolescents and young adults aged 15–24 years in Kenya. Using national TB data from **TIBU** alongside qualitative interviews and participatory workshops with young people, the study found important losses across the care cascade. A large proportion of health facility attendees were not screened for TB, and many presumptive TB cases were not tested. Qualitative findings showed that pharmacies were often the first

point of care for young people, while delays in diagnosis were common because symptoms were initially treated as other illnesses such as pneumonia, flu, asthma, or tonsillitis. Young men often delayed seeking care because of work responsibilities, while young women faced caregiving burdens, stigma, and family pressures. Supportive healthcare workers, flexible clinic schedules, confidentiality, and family support emerged as important enablers of treatment completion.

02

The **second study** focused on TB-related stigma among people receiving TB treatment in Kenya. Conducted across multiple counties using a structured TB stigma index, the survey found that stigma remains a major barrier to care. Community stigma was the most commonly reported form, followed by family-related and healthcare-related stigma. Many people affected

by TB concealed their diagnosis because of fear of discrimination, while some delayed seeking care because they anticipated negative reactions from others. Stigma was particularly influenced by fear of infection, misconceptions linking TB with HIV, and weak social support systems. These findings highlight the continued need for stigma reduction interventions within communities and health facilities.



03

The third study, a national Community, Rights and Gender Assessment, explored how social, legal, and gender-related factors influence access to TB services. Conducted across 11 counties using mixed methods, the assessment found that financial barriers, transport costs, and limited social protection continue to affect treatment access and continuity. Many respondents reported limited prior knowledge about TB before diagnosis, while a substantial proportion

were unaware of legal protections related to confidentiality, isolation, and non-discrimination. Gender norms influenced TB care differently for men and women: men often delayed care because of work-related responsibilities, while women in some settings experienced delayed access because of household decision-making dynamics. The study also highlighted additional barriers faced by vulnerable populations, including migrants, people in remote settings, and gender-diverse groups.

04

The fourth study assessed tuberculosis knowledge, attitudes, and practices in the general population in Kenya through a national household survey. Although awareness of TB was high, important knowledge gaps remained. Most respondents had heard of TB, but only a small proportion correctly identified bacteria as the cause of the disease. Misconceptions about transmission were common, including beliefs that TB spreads through sharing food or utensils. While cough was widely recognized as a symptom, fewer people identified other important symptoms such as weight loss, night sweats, and chest pain. Attitudes were generally positive, although stigma remained evident in a significant

proportion of respondents. Encouragingly, most people reported that they would seek care at a health facility if they developed TB symptoms, though awareness of drug-resistant TB remained low.

Taken together, these studies show that while Kenya has made progress in strengthening TB services, important barriers remain across service delivery, community perceptions, social protection, rights, and communication. The evidence points to the need for stronger age- and sex-disaggregated TB data, expanded community awareness, stigma reduction, improved patient-centred care, and greater integration of youth-, gender-, and rights-responsive approaches within TB programming.



TB Care Cascade Among Adolescents and Young Adults in Kenya

Understanding barriers, treatment experiences, and opportunities for youth- and gender-responsive TB care



BACKGROUND

Tuberculosis (TB) remains highly prevalent among young people aged 15–24 years in sub-Saharan Africa, despite continued investment in TB prevention, treatment, and care. Males carry a disproportionately higher TB burden; however, the reasons underlying this gender disparity remain incompletely understood. While biological factors may contribute, gender-related barriers across the TB care cascade are likely to play an important role in worsening this inequity.

The high burden of TB among young people reflects increased risks of infection, re-infection, and ongoing transmission within communities. Early case finding and timely initiation of treatment are essential to reduce both health and socio-economic consequences of TB. Addressing these challenges requires a deeper understanding of the barriers and facilitators influencing progression across the TB care cascade, particularly among vulnerable groups. This study, therefore, examined barriers and enablers to successful completion of the TB care cascade among young people in Kenya in order to inform age- and gender-responsive interventions.

01

WHAT DID WE DO?

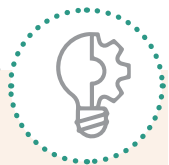
We conducted an observational cross-sectional study using a mixed-methods approach.

The quantitative component involved analysis of national TB data captured in TIBU across all 47 counties to map patient pathways and identify points of loss along the TB care cascade.

The qualitative component included in-depth interviews with young people who had disengaged from TB care, as well as participatory workshops with young people who completed treatment. During the workshops, participants used visual methods, including *Rivers of Life*, to describe their care experiences. They also identified common barriers and, through *Stepping Stones*, proposed practical solutions for improving TB services for young people.

In addition, roundtable discussions were held with TB stakeholders, including healthcare workers, community leaders, community health promoters, young TB champions, and sub-county and county health leadership. Study findings were shared to obtain feedback and jointly identify priority interventions tailored to young people aged 15–24 years in Kenya.

02



WHAT WERE THE RESULTS?

Gaps Along the TB Care Cascade from TIBU Data

Analysis of active case-finding data from TIBU for the period 2017–2022 showed that screening data are currently available only in aggregate form, limiting age- and sex-disaggregated analysis.

- **Screening and testing gaps:** Here, 73% of health facility attendees were not screened for TB, while 42% of presumptive TB cases were not tested.
- **Linkage to treatment gap:** More individuals were documented as receiving TB treatment than those diagnosed through GeneXpert, suggesting data inconsistencies between diagnostic and treatment records.

03



- **Retention on treatment:** Females had slightly higher treatment success than males. Patients aged 15–19 years recorded the highest treatment success rate at 92%, compared with 89% among those aged 20–24 years.

Findings from Qualitative Studies (In-depth Interviews and Participatory Workshops)

Delayed and Confusing Pathways to Diagnosis

- Pharmacies were commonly the first point of care for young people with TB symptoms.
- In health facilities, symptoms were frequently misdiagnosed as pneumonia, flu, tonsillitis, asthma, or pregnancy, resulting in delays of 2–3 months before TB diagnosis.
- Poor communication around diagnosis, including initiation of treatment without clear explanations, contributed to confusion and frustration.
- Multiple referrals, repeated facility visits, and costs of incorrect treatment created financial and emotional strain.

Psychosocial and Structural Barriers

- Rigid referral systems and service delivery barriers reduced trust in care.
- **For young men:** Work demands, masculinity norms, incarceration, and homelessness were major barriers. Many delayed seeking care until illness interfered with work.
- **For young women:** Caregiving responsibilities, family conflict, stigma, and misconceptions such as spiritual causes of TB affected care seeking and adherence.

- **Across both groups:** long waiting times, travel fatigue, rigid clinic hours, treatment side effects, stigma, discrimination, financial strain, and poor communication affected treatment continuity.

Enablers of Successful Treatment

- Supportive and non-judgmental healthcare workers improved treatment adherence.
- Youth-friendly staff, social worker support, and occasional transport reimbursement facilitated clinic attendance.
- Flexible drug refill schedules, family support, peer support, and confidentiality in care (such as discreet medicine packaging) promoted adherence.
- Counselling that addressed fears and treatment concerns encouraged continued engagement in care.

Why Some Adolescents and Young Adults Disengage

- **Young men:** Work responsibilities were the main reason for treatment interruption. Others cited incarceration and homelessness.
- **Young women:** Emotional stress at home, stigma, financial difficulties, and household responsibilities were major contributors. Compassionate follow-up by nurses or peer educators often supported re-engagement.
- **Across both groups:** Side effects, long queues, distance to facilities, healthcare worker strikes, homelessness, and broader disruptions such as political demonstrations affected continuity of care.

WHAT ARE THE RECOMMENDATIONS?



1. Strengthening TB Data Quality and Use

- Strengthen age- and sex-disaggregation across the full TB care cascade, beyond treatment outcomes.
- Address discrepancies between laboratory and treatment data, including improved alignment with TIBU Lite.



2. Finding the “Missing” Cases Outside TB Sites

- Expand TB training and mentorship to non-TB treatment sites and private health outlets.
- Leverage chemists and pharmacies as referral points for individuals with respiratory symptoms.

- Place TB awareness and referral materials in chemists and pharmacies to support early care seeking.



3. Patient-Centred Care: Youth- and Gender-Responsive TB Programming

- Position adolescents, young people, and gender-responsive approaches as explicit priorities within TB programming.
- Use findings to inform policy discussions, National Strategic Plan review, Global Fund applications, future surveys, and implementation of the TB in school policy.



TB-Related Stigma Among People with Tuberculosis in Kenya



BACKGROUND

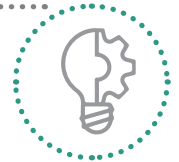
01

- Tuberculosis-related stigma remains a major barrier to early diagnosis, treatment adherence, and treatment completion in Kenya. Stigma affects people with TB at multiple levels, within families, communities, healthcare settings, and workplaces, often leading to delayed disclosure, social isolation, and reduced access to care.

Although stigma has long been recognized as a challenge in TB control, Kenya had not previously conducted a national survey to measure the extent and dimensions of TB-related stigma. This study was undertaken to generate national evidence to guide stigma reduction interventions within TB programming.

WHAT DID WE DO?

We conducted a national cross-sectional TB stigma survey in 2023 across 180 health facilities in 11 counties in Kenya.



02

- Participants were people with TB aged 15 years and above who were actively on treatment.
- A total of 367 people with TB were interviewed using a structured TB stigma index tool.
- The survey assessed stigma across five dimensions: self-stigma, family stigma, community stigma, healthcare-related stigma, and workplace stigma.
- Data were analysed to identify the most common forms of stigma and factors associated with stigma.



WHAT WERE THE RESULTS?

Major Dimensions of TB Stigma

- 68% of participants reported stigma from the community, making community stigma the most common form.
- 52% experienced stigma within the family.
- 51% reported stigma within healthcare settings.
- 49% experienced self-stigma.
- 11% reported workplace stigma affecting access to TB services.

03

How Stigma Affected Care

- Many people with TB concealed their diagnosis due to fear of discrimination.
- Some delayed seeking care or avoided TB services because of anticipated stigma.
- Community stigma was the strongest predictor of care disruption.

Groups More Affected

- Stigma was more commonly reported among adults aged 25–44 years.
- Men reported stigma more frequently than women.

- Newly diagnosed individuals and those in the intensive treatment phase experienced higher stigma than those who were previously treated, and also those in the continuation phase.

Main Drivers of Stigma

- Fear of infection in communities
- Misconceptions linking TB with HIV
- Gossip and social distancing
- Negative attitudes in some healthcare settings
- Weak social support systems

WHAT ARE THE RECOMMENDATIONS?

04



Strengthening Community Awareness

- Intensify community education to address misconceptions about TB transmission and treatment.
- Normalize TB as a curable disease through sustained public awareness campaigns.



Reducing Stigma in Health Services

- Strengthen stigma reduction training for healthcare workers.
- Promote respectful, confidential, and non-discriminatory TB care.



Strengthening Psychosocial Support

- Integrate psychosocial support into TB care, especially during the intensive phase of treatment.
- Strengthen peer and family support mechanisms for people with TB.



Policy and Programme Action

- Integrate stigma monitoring into routine TB programme reviews.
- Use stigma findings to guide patient-centred TB service delivery and community engagement strategies.



Community, Rights and Gender Assessment (CRGA) in Kenya



BACKGROUND

- Tuberculosis control is influenced not only by access to diagnosis and treatment, but also by how community rights, gender dynamics, and the legal environment shape people's ability to seek and complete care. Certain groups, including women, key and vulnerable populations, and people living in remote settings, face additional barriers related to stigma, financial constraints, limited social protection, and unequal access to information.

The Community, Rights and Gender Assessment (CRGA) was conducted to generate evidence on how TB services in Kenya respond to issues of rights, gender, and equity, and to identify actions needed to strengthen a people-centred and rights-based TB response.

01

WHAT DID WE DO?

We conducted a national mixed-methods assessment across 11 counties involving 1,544 participants drawn from populations affected by TB and key stakeholders.

The assessment included:

- Structured questionnaires among people affected by TB
- Focus group discussions with people with TB, including key and vulnerable populations

- In-depth interviews with people with TB, family members, students, and community leaders
- Key informant interviews with healthcare workers, prison officers, county TB coordinators, and civil society representatives

The study assessed TB services across key domains of availability, accessibility, acceptability, quality, gender responsiveness, legal environment, and community rights.



02



WHAT WERE THE RESULTS?

03

Access and Support Gaps

- 37% of respondents reported financial barriers to accessing TB services, mainly transport costs.
- 29% reported TB services were not within safe physical reach, particularly in remote counties.
- 37% did not receive treatment support such as nutritional, financial, or social assistance during treatment.
- 70% faced barriers in accessing social protection and other benefits.

Information and Rights Gaps

- 60% reported having no information about TB before diagnosis.
- 35% reported unequal access to TB information for indigenous populations, migrants, and internally displaced persons.
- 40% hid their TB diagnosis from certain individuals, mainly employers, due to fear of losing jobs and stigma.
- 93% were unaware of legal standards governing isolation, quarantine, or detention of people with TB.

Gender and Vulnerability Findings

- Gender norms influenced TB care-seeking and treatment adherence differently for men and women.
- Patriarchal structures limited women's timely access to TB services in some counties.
- Work responsibilities delayed care-seeking among men.
- Transgender and LGBTQI populations reported additional barriers related to stigma and access to friendly services.

Quality of Care

- 98% reported TB diagnosis and treatment services were generally of good quality.
- However, delayed care-seeking remained common, with 17% seeking care more than three months after symptom onset.
- Treatment side effects were frequently reported and affected treatment acceptability.



WHAT ARE THE RECOMMENDATIONS?

04



Strengthening Rights-Based TB Care

- Integrate community rights education into TB services so patients understand their legal protections and entitlements.
- Improve awareness of legal standards related to isolation, confidentiality, and non-discrimination.



Improving Social and Financial Protection

- Strengthen social support for people on TB treatment, including transport, nutrition, and psychosocial support.
- Link TB patients to social protection systems to reduce treatment interruption caused by economic hardship.



Strengthening Gender-Responsive TB Programming

- Integrate gender-sensitive approaches into TB prevention, diagnosis, and treatment.
- Address specific barriers affecting women, men, and gender-diverse populations.



Improving Access to Information

- Expand TB education using local languages and community-friendly approaches.
- Target underserved populations including remote communities, migrants, and vulnerable groups.

Tuberculosis Knowledge, Attitudes and Practices (KAP) Survey in Kenya



BACKGROUND

Tuberculosis remains a major public health concern in Kenya, with a substantial proportion of expected TB cases still missed each year. Limited knowledge, misconceptions, stigma, and delayed health-seeking behaviour continue to affect early diagnosis and treatment uptake.

The national TB Knowledge, Attitudes and Practices (KAP) survey was conducted to assess how the general population understands TB, perceives people affected by TB, and responds to TB symptoms, to guide advocacy, communication, and community engagement strategies.

01

WHAT DID WE DO?

We conducted a national cross-sectional household survey in 24 counties, covering 4,171 individuals from 2,783 households.

- Participants were aged 15 years and above and selected through multistage cluster sampling.
- Data were collected through face-to-face household interviews using structured questionnaires.

- The survey assessed TB-related knowledge, attitudes, practices, and preferred communication channels.
- Quantitative data were analyzed to identify differences across counties, sex, age groups, education level, and residence.



02

WHAT WERE THE RESULTS?

TB Knowledge Gaps

- 95% of respondents had heard about TB.
- However, only 23% correctly identified bacteria as the cause of TB.
- 36% did not know the cause of TB.
- 76% correctly identified airborne transmission through coughing or sneezing.
- 40% still believed TB can spread through sharing food or utensils.

Knowledge of Symptoms and Treatment

- 90% identified cough as a TB symptom.
- Recognition of other symptoms was much lower: weight loss (32%), chest pain (26%), night sweats (21%), and fever (19%).
- Only 11% correctly recognized that a cough of any duration can raise suspicion of TB.
- Knowledge of drug-resistant TB remained low (33% awareness).

Attitudes Towards TB

- 76% were willing to work with a person with TB.
- However, 24% still held stigmatizing attitudes.
- Misconceptions linking TB with HIV remained common.

TB Practices and Health-Seeking Behaviour

- 96% reported they would seek care at a health facility when experiencing TB symptoms.
- 64% were aware that sputum testing is used to diagnose TB.
- Younger people aged 15–24 years showed the most positive attitudes and best TB-related practices.

03



Population Groups with Lower Knowledge

- ➔ Lower TB knowledge was observed among:
 - older adults (65 years and above)
 - rural populations
 - individuals with no formal education

WHAT ARE THE RECOMMENDATIONS?



04



Strengthening TB Communication

- ➔ Intensify public education on TB causes, symptoms, transmission, and treatment using simple and locally understood language.
- ➔ Address persistent misconceptions, especially around TB transmission and causes.



Targeting Underserved Populations

- ➔ Prioritize rural communities, older adults, and low-literacy populations for TB awareness campaigns.
- ➔ Use local communication channels that communities trust most.



Reducing Stigma

- ➔ Strengthen community messaging that TB is preventable, treatable, and curable.
- ➔ Integrate anti-stigma messages into routine TB advocacy.



Improving Early Care Seeking

- ➔ Promote recognition of early TB symptoms and encourage prompt care-seeking.
- ➔ Strengthen lower-level health facilities as trusted entry points for TB services.



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